HCP 1826 (pc_hcp_507): a cipher postcard from Carrie to Fannie J. Roberts

Human solver: Tomáš Pavuk

Status: deciphered in full (300 of 300 cipher signs); independent review 1 verdict REVISE, one word corrected, all findings addressed. Report dated 2026-10-30.

Origin and context

The cryptogram is record 1826 of the HC Portal (https://crypto.hcportal.eu/dashboard/cryptograms/1826), portal name pc_hcp_507, part of the HCPortal collection of cryptographic postcards. When it was captured on 2026-10-30 the portal listed it as **Not solved**, tagged Postcard, HCPortal collection, category Substitution, with date and language unknown, location Europe, sender Unknown and recipient "Jannie f. Roberts". No transcription or solution was attached.

The card itself is American. The address, in plain writing, reads **Miss Fannie J. Roberts, Bloomville, Delaware Co., N.Y., R.D. #1**, so the portal's "Jannie f." is a misreading and "Europe" does not fit the addressee. The stamp is a green 1-cent Washington. The postmark is only partly legible: "JU", a third letter that may be L or N, and "191" with the last digit lost. So the card was mailed in June or July of a year in the 1910s. The text fits that, since "the last of Aug." still lies ahead.

The picture side shows a bluebird and the printed verse "I send my best wishes flying to you side by side with the emblem of Happiness." A pencil note on it reads best as Carrie Craig; the first name is clear and the surname is less so.



Address side with the cipher text: the side block at 90 degrees and the upside-down block above it (source/address-large.jpg)



Picture side with the printed verse and the pencil name at the left edge (source/picture-large.jpg)

All cipher text is on the message half of the address side. Neither side carries any other cipher text.

Method

The transcription is manual. The signs were read from enlarged crops of the original scan (work/hires/pc_hcp_507_adr.JPG, 1785 × 1175 pixels). The cipher writes numbers, so the first step was to see that the number groups stand for letters, with dotted and bracketed forms as separate signs.

The break came from cribs. A recurring group 4 14 2' gave "the", 4 14 X 4 gave "that" and 0' 1' 4 14 gave "with". Then "on the train that" in row S03 and "your card received" in row S01 gave most of the rest. The left and right brackets turned out to separate o and u, and circles turned out to mark capitals. After that, the consonants showed a regular pattern, which predicted the remaining ones.

The reading order is not obvious. The side block runs at 90 degrees from the divider leftwards (rows S01–S12). It continues in the upside-down block above it, from the card centre out to the top edge (rows T1–T6), and ends in the signature. "Let me" at the end of S12 runs straight into "knew if you" in T1.

In review, a separate agent typed all 18 rows from its own crops, without word division and in its own notation, and decoded them with its own script. It agreed with the solver on all signs except one word (see Review summary).

Key

Monoalphabetic substitution in English with a hand-made numeric alphabet. Consonants are numbered downwards from b = 19, skipping vowels. Vowels, w and y have special signs. A circle around a sign marks a capital.

Plain	Sign	Plain	Sign	Plain	Sign	Plain	Sign
a	X (crossed Z-like sign)	b	19	c	18	d	17
e	2 with raised dot	f	16	g	15	h	14
i	1 with raised dot	j	13	k	12	l	11
m	10	n	9	o	0 in left bracket	p	8
r	6	s	5	t	4	u	0 in right bracket
v	3	w	0 with raised dot	y	0 plain		

The letters q, x and z do not occur. The sequence would give q = 7, but 7 never stands alone on the card. The dots after 1, 2 and 0 are part of the letter, not punctuation. The time 1:48 is written as a plain numeral. Capitals seen: circled 19 = B, 1 = I, X = A, 4 = T, 11 = L, 18 = C.

Plaintext

Transcription, row by row. X is the crossed sign, 2' 1' 0' are dotted forms, [0 and 0] are the bracketed forms, (n) is a circled sign, | is the editor's word division, and {1:48} is a plain numeral.

```
S01 0 [0 0] 6 | 18 X 6 17 | 6 2' 18 2' 1' 3 2' 17
S02 0' X 5 | 8 11 X 9 1' 9 15 | [0 9 | 15 [0 1' 9 15
S03 [0 9 | 4 14 2' | 4 6 X 1' 9 | 4 14 X 4
S04 15 2' 4 5 | 4 [0 | (19) | X 4 | {1:48}
S05 0' 2' 17 9 2' 5 17 X 0 | 19 0] 4 | 1' 16 | 0 [0 0]
S06 17 [0 | 9 [0 4 | 0' X 9 4 | 10 2' | 4 [0 | 18 [0 10 2'
S07 4 14 2' 9 | 13 0] 5 4 | 5 X 0 | 5 [0 | X 5 | 16 [0 6
S08 4 14 2' | [0 4 14 2' 6 | 3 1' 5 1' 4 | (1) | 17 [0 | 9 [0 4
S09 0' X 9 4 | 4 [0 | 15 [0 | 9 [0 0' | 0] 9 4 1' 11 | 4 14 2'
S10 11 X 5 4 | [0 16 | (X) 0] 15 . | 4 14 2' 9 | (4) . | 18 X 9
S11 15 [0 | 0' 1' 4 14 | 10 2' | X 9 17 | 4 14 2' 6 2' | 1' 5
S12 X 9 [0 4 14 2' | 6 2' X 5 [0 9 . | (11) 2' 4 | 10 2'
T1 12 9 2' 0' | 1' 16 | 0 [0 0]
T2 0' X 9 4 | 10 2' | 4 [0 | 18 [0 10 2' | 0' 2' 17 . | [0 6
T3 0' [0 0] 11 17 | 6 X 4 14 2' 6 | (1) | 0' [0 0] 11 17
T4 18 [0 10 2' | 5 [0 10 2' | [0 4 14 2' 6 | 4 1' 10 2'
T5 (1) | 13 0] 5 4 | X 5 | 5 [0 0] 9 | 14 2' 11 8 | 0 [0 0] | 0' 14 X 4
T6 (1) | 18 X 9 | X 9 0 | 0' X 0 . | 11 [0 3 2' | (18) X 6 6 1' 2' .
```

Literal plaintext, in the writer's spelling:

```
S01  your card received
S02  was planing on going
S03  on the train that
S04  gets to B at 1:48
S05  wednesday but if you
S06  do not want me to come
S07  then just say so as for
S08  the other visit I do not
S09  want to go now until the
S10  last of Aug. then T. can
S11  go with me and there is
S12  anothe reason. Let me
T1   knew if you
T2   want me to come wed. or
T3   would rather I would
T4   come some other time
T5   I just as soun help you what
T6   I can any way. love Carrie.
```

Normalized English (punctuation and capitals are editorial):

Your card received. Was planning on going on the train that gets to B[loomville?] at 1:48 Wednesday, but if you do not want me to come then just say so. As for the other visit, I do not want to go now until the last of Aug.; then T. can go with me, and there is another reason. Let me know if you want me to come Wed. or would rather I would come some other time. I [would] just as soon help you what I can anyway. Love, Carrie.

Slovak translation:

Tvoju kartu som dostala. Plánovala som ísť vlakom, ktorý prichádza do B[loomville?] v stredu o 1:48, ale ak nechceš, aby som prišla, tak to jednoducho povedz. Čo sa týka tej druhej návštevy, nechcem ísť teraz, až koncom augusta; potom môže ísť so mnou T., a je tu ešte iný dôvod. Daj mi vedieť, či chceš, aby som prišla v stredu, alebo by si radšej chcela, aby som prišla inokedy. Rada by som ti pomohla, ako len budem môcť. S láskou, Carrie.

Reproduction

From the repository root:

```
python3 -I ciphers/hcp-1826/work/decode.py
```

The script uses only the Python standard library. It checks the SHA-256 hashes of the two source scans and the two originals, decodes work/transcription.txt with the key, and prints the 18 rows and the literal plaintext. It reports 300 signs, 29 distinct forms and no unkeyed token. It does not read the images; the transcription step is manual and was checked independently in review.

Uncertainties

- **"what" at the end of T5.** The word is cramped in the corner of the card. The reading 0' 14 X 4 is strong but not certain; see the Review summary. No higher scan exists.
- **The circled C of Carrie.** The sign is cramped at the card edge and the arc over the 8 is thin. A reader could take it for a 0 beside an 8. The word and the pencil name support the circle. A small dot above this sign, apart from the dot of "love", may be a stray or a full stop; it changes no letter.
- **The writer's slips, kept in the literal text.** "planing"; "knew" for know (the sign is clearly 2 with a dot); "soun" for soon (the second box is closed on the right, as in "you"); "anothe reason" (only one 6 between the words, so a sign is shared or an r was dropped).
- **o or u in three places.** In S09 "until", S10 "of" and S12 "reason" the box looks closed on both sides or is faint on the printed imprint. These were settled by the word.
- **Faded 5 in S10 "Aug."** The sign has the separate top bar of the writer's 5 and is read as 15 = g.
- **B and T.** are bare initials. B may be Bloomville, the recipient's town, but that is a guess. T. is unknown.
- **Date and identity.** The exact date is not legible. The surname Craig is outside the cipher and stays uncertain. The relationship between Carrie and Fannie is not known, though "help you" suggests a family or close friend.
- **Editorial choices.** Word divisions and all punctuation except the writer's full stops are editorial.
- **Prior publication.** The portal attaches no solution. The AI Decipherment Register had no record of this card on 2026-10-30. The reviewer searched only this repository for related cards, and no wider literature search was made.

Review summary

Reviewer: a separate Claude Opus 5.5 agent, round 1, verdict **REVISE**. The reviewer ran the reproduction command and read all 18 rows sign by sign on contrast-enhanced crops. Its own decoder gave 300 letters against the solver's 300, with one difference. It confirmed the key over the whole card, the reading order, and the solver's disclosed slips.

- **Major, fixed: the last word of T5 is "what", not "with".** The solver had first read 0' 1' 4 14. On the card, the dot belongs to the 0', and no dot follows the 1, although the other 14 i on the card all have one. The fourth sign is much wider than a 1 and has a top bar, which makes it the X. The ending then matches "that" in S03. The first reading had also forced a split into "help you with [it]. I can anyway", while the corrected one gives a single plain sentence. The solver rechecked the crop and agreed. The total stays at 300 signs.
- **Minor, fixed:** the claim that T6 was the only reading resting on the word itself was reworded.
- **Minor, fixed:** the stray dot above the circled C is now disclosed.
- **Minor, fixed:** the dating evidence now gives the third postmark letter and the agreement with "the last of Aug.", and the pencil surname now reads Craig, marked uncertain.



Top: the end of T5 from the card corner, rotated, showing 0' 1 4 X 4. Middle: "that" from S03 with the same ending. Bottom: "with" from S11, with both dots (review/work/crops/with_compare.png)

The key and all other signs are unchanged. Transcription, analysis, decipherment and this report were done by Claude Opus 5.5; no earlier work on this card exists.

Sources

- HC Portal record 1826, General, Description and Attachments tabs, captured 2026-10-30: https://crypto.hcportal.eu/dashboard/cryptograms/1826 (source/metadata.md).
- Picture side: https://api.hcportal.eu/media/5538/conversions/pc_hcp_507_pic-large.jpg (source/picture-large.jpg); original https://api.hcportal.eu/media/5538/pc_hcp_507_pic.JPG (work/hires/).
- Address side: https://api.hcportal.eu/media/5539/conversions/pc_hcp_507_adr-large.jpg (source/address-large.jpg); original https://api.hcportal.eu/media/5539/pc_hcp_507_adr.JPG (work/hires/).
- AI Decipherment Register (https://ai.hcportal.eu/register/), checked 2026-10-30.